CS-1101
PROGRAMMING FUNDAMENTALS

Lab 10

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Lab Contents

- Arrays

Lab Objectives

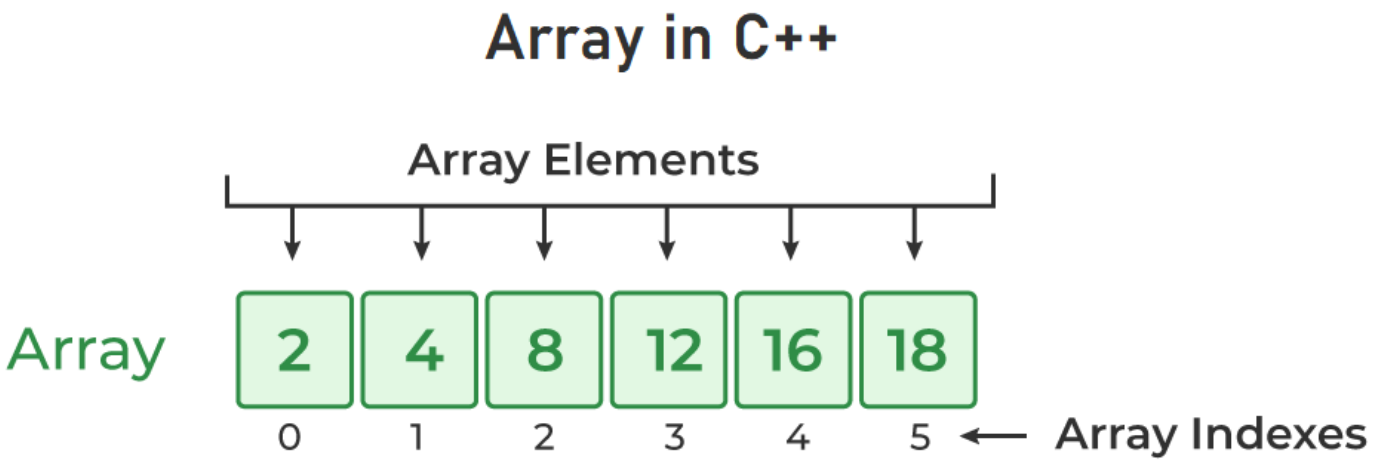
- The objective of this lab is to understand the concept of arrays in C++.
- This includes practicing tasks such as initializing arrays, traversing them using loops, finding specific elements, performing calculations, and understanding their properties like size and indexing.

Understanding the Problem

- **For example**, we have to store the marks of 5 students then we can easily store them by creating 5 different variables but what if we want to store marks of 100 students or say 500 students then it becomes very challenging to create that numbers of variable and manage them.
- Now, arrays come into the picture that can do it easily by just creating an array of the required size.

What is an array

- An array is a data structure that is used to store multiple values of similar data types in a contiguous memory location.

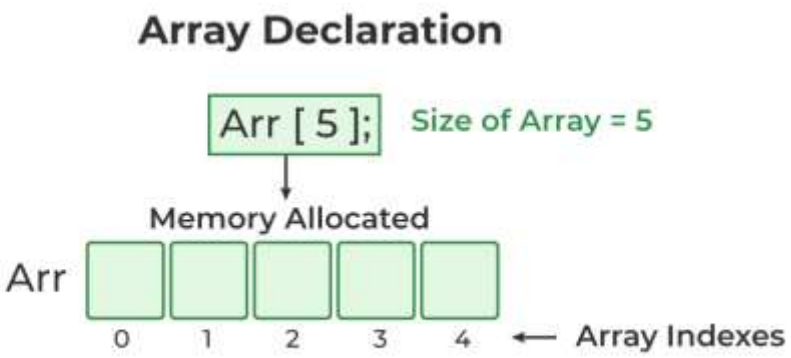


Properties of Array

- An Array is a collection of data of the **same data type**, stored at a **contiguous memory location**.
- Indexing of an array starts from 0. It means the first element is stored at the **0th index**, the second at 1st, and so on.
- Elements of an array can be accessed using their **indices**.
- Once an array is declared its **size remains constant** throughout the program.
- An array can have **multiple dimensions**.
- The size of the array in bytes can be determined by the **sizeof** operator using which we can also find the number of elements in the array.

Array Declaration in C ++

- data_type array_name[Size_of_array];
- **Example** : int arr[5];
- **int**: It is the type of data to be stored in the array. We can also use other data types such as char, float, and double.
- **arr**: It is the name of the array.
- **5**: It is the size of the array which means only 5 elements can be stored in the array



Initialization of Array in C++

- **Initialize Array with Values in C++**
  - int arr[5] = {1, 2, 3, 4, 5};
- **Initialize Array with Values and without Size in C++**
  - int arr[] = {1, 2, 3, 4, 5};
- **Initialize Array after Declaration (Using Loops)**
  - for (int i = 0; i < N; i++) {

    arr[i] = value;

    }
- **Initialize an array partially in C++**
  - int partialArray[5] = {1, 2};
- **Initialize the array with zero in C++**
  - int zero_array[5] = {0};

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- **Initialize the array with zero in C++**
  - int zero_array[5] = {0};

Accessing an Element of Array in C++

- Elements of an array can be accessed by specifying the name of the array, then the index of the element enclosed in the array subscript operator []. For example, arr[i].

```cpp
//  C++ Program to Illustrate How to Access Array Elements
#include <iostream>
using namespace std;

int main()
{

    int arr[3];

    // Inserting elements in an array
    arr[0] = 10;
    arr[1] = 20;
    arr[2] = 30;

    // Accessing and printing elements of the array
    cout << "arr[0]: " << arr[0] << endl;
    cout << "arr[1]: " << arr[1] << endl;
    cout << "arr[2]: " << arr[2] << endl;

    return 0;
}
```

Output

```
arr[0]: 10
arr[1]: 20
arr[2]: 30
```

Updating an Element of Array in C++

- To update an element in an array, we can use the index which we want to update enclosed within the array subscript operator and assign the new value.
- arr[i] = new_value;

Traverse an Array in C++

- We can traverse over the array with the help of a loop using indexing in C++.

```cpp
// C++ Program to Illustrate How to Traverse an Array
#include <iostream>
using namespace std;

int main()
{

    // Initialize the array
    int table_of_two[10]
        = { 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 };

    // Traverse the array using for loop
    for (int i = 0; i < 10; i++) {
        // Print the array elements using indexing
        cout << table_of_two[i] << " ";
    }

    return 0;
}
```

Output

```
2 4 6 8 10 12 14 16 18 20
```

Size of an Array in C++

- In C++, we do not have the length function as in Java to find array size but we can calculate the size of an array using sizeof() operator trick.
- First, we find the size occupied by the whole array in the memory and then divide it by the size of the type of element stored in the array. This will give us the number of elements stored in the array.

Array out of bound

- If we declare an array of size 10, then the array will contain elements from index 0 to 9.
- However, if we try to access the element at index 10 or more than 10, it will result in an undefined behavior.
- This type of error is likely when we use a normal for loop and we access the array element using the [] operator.

Tasks of Arrays

- **Program to Initialize and Print an Array:**

Write a program that initializes an array of five integers, then traverses and prints the array elements.

- **Program to Print an Array in Reverse Order:**

Write a program that asks the user to input an array of N integers. Then print the array elements in reverse order.

- **Program to Calculate Sum and Average of Array Elements**:

Write a program that initializes an array of integers. Compute the sum and average of all the elements in the array and display the results.

- **Program to Find the Length of an Array:**

Write a program that initializes an array of integers and then calculates and prints the size (length) of the array.

- **Program to Find the Largest number in an Array:**

Write a program that prompts the user to input an array of N integers and then determines and displays the largest number in the array.

References

- https://www.geeksforgeeks.org/cpp-arrays/